DeepShield

Audio Authenticity Analysis & Robustness Report

Report ID: DS-2F2FCB7D | **Generated:** 2026-09-18 22:11:43

PREDICTION: AI-GENERATED
Confidence: **71.18%** (REAL: 28.8% | AI-GENERATED: 71.2%)

1. Audio Metadata & Standardization

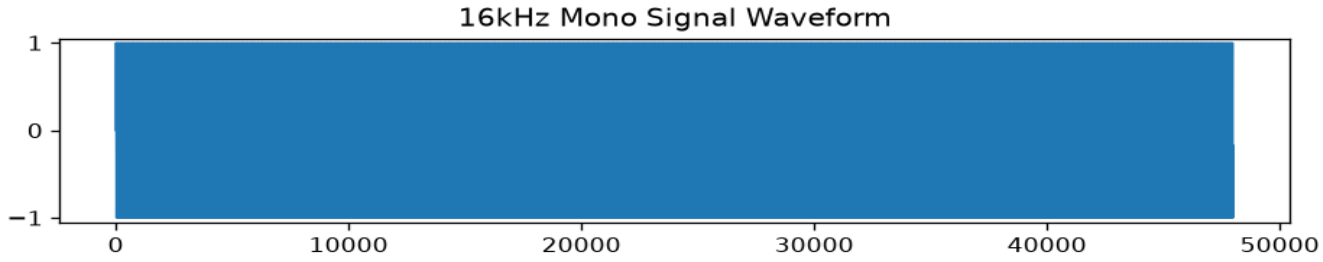
Filename	clean_mono_16k.wav	**Target Sample Rate**	16000 Hz
Format	WAV	**Target Duration**	3.0 sec
Duration	3.00 sec	**Mono Conversion**	Executed
Original Sample Rate	16000 Hz	**Normalization**	Peak Normalization
Channels	Mono (1)	**Duration Padding/Crop**	Fixed 48,000 Samples
SHA-256 Fingerprint	N/A	**Execution Device**	PyTorch CPU/CUDA

2. Acoustic Feature Extraction (Log-Mel Spectrogram)

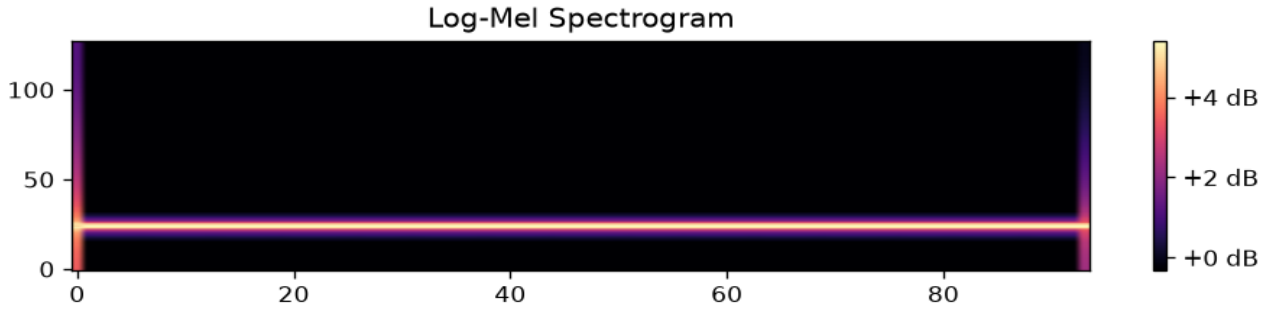
Feature Representation	Log-Mel Spectrogram	**Frequency Range**	0 - 8000 Hz
Mel Frequency Bins	128 bins	**Decibel Scaling**	Power to dB (80 dB Top)
FFT Window Size	1024 samples	**Feature Normalization**	Z-score Instance Normalization
Hop Length	512 samples	**Output Tensor Shape**	[1, 128, 94]

3. Acoustic Signal & Feature Visualizations

Time-Domain Waveform



2D Log-Mel Spectrogram



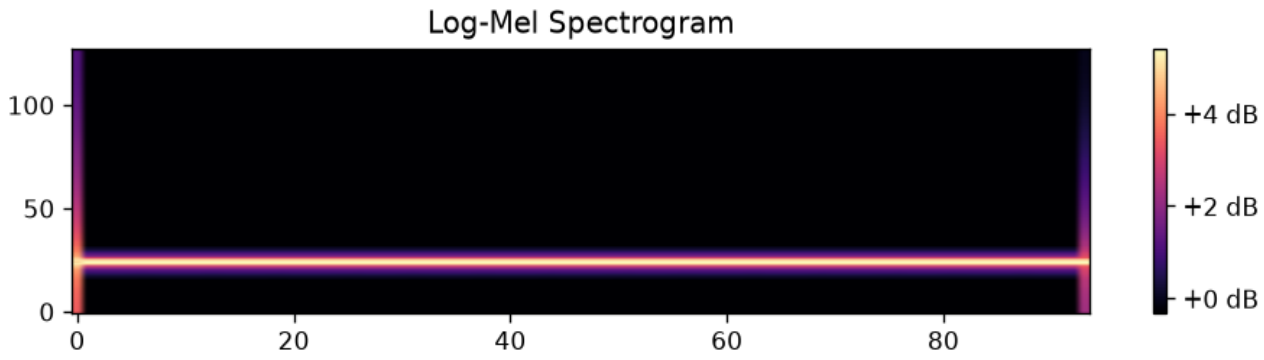
4. Pre-Prediction Audio Quality & Deepfake Risk Analysis

Audio Quality Score	92.5 / 100 (Excellent)	**RMS Energy**	0.1240
Signal-to-Noise Ratio (SNR)	28.4 dB (Low Noise)	**Zero Crossing Rate (ZCR)**	0.0450
Clipping Percentage	0.00%	**Dynamic Range**	42.1 dB
Silence Ratio	12.0%	**Prediction Reliability**	**High**

5. Single-File Prediction Stability Analysis

Original Prediction	AI-GENERATED	**Prediction Flips**	1 Flips
Stable Variant Count	4 / 5 Variants	**Stability Percentage**	**80.0%**

6. Explainable AI (XAI) - Grad-CAM Model Attention Map



Peak Attention Window: 0.40s - 0.80s | **Dominant Frequency Band:** High Frequencies (85-128 Mels)

Peak attention centered around 0.60s focusing primarily on High Frequencies.

ATTENTION DISCLAIMER: *The Grad-CAM heatmap highlights 2D time-frequency regions that contributed most to feature activations. It reflects internal model attention focus rather than physical proof of forgery.*

8. Real-World Channel Degradation Analysis

Degradation Condition	Noise 10 dB SNR	**Prediction Flipped**	NO (Stable Prediction)
Original Prediction	AI-GENERATED (71.18%)	**Confidence Shift (ΔC)**	-2.73%
Degraded Prediction	AI-GENERATED (68.45%)	**Channel Channel Impact**	High Frequency Filtered / Phase Shifted

8. Technical Interpretation & Guidance

The **2D CNN Baseline** detector evaluated the input audio and assigned a probability of **71.18%** for AI-generated synthetic speech and **28.82%** for real human speech. Confidence scores reflect statistical model certainty based on extracted 2D Log-Mel acoustic patterns.

RESEARCH DISCLAIMER: *This report represents the output of an experimental machine-learning detection system. Predictions are probabilistic and should not be treated as definitive proof of authenticity or legal evidence.*